EXAM SHIELD

Golang Question Bank

Concurrency, error handling, types and tooling in Go.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 20 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Golang

Concurrency, error handling, types and tooling in Go.

1. Which keyword starts a new goroutine?

A. go
B. spawn
C. async
D. thread

Answer: A

2. What is the value of a map declared with var m map[string]int before it is initialised?

A. A map with one zero entry
B. An empty map ready for writes
C. It does not compile
D. nil

Answer: D

3. What happens when a program writes to a nil map in Go?

A. The map is created automatically
B. The write is silently ignored
C. A compile-time error
D. A runtime panic

Answer: D

4. Which statement schedules a function call to run when the surrounding function returns?

A. later
B. defer
C. await
D. finally

Answer: B

5. How does idiomatic Go report a recoverable failure from a function?

A. By throwing an exception
B. By returning an error value as the last result
C. By setting a global errno variable
D. By calling panic in every case

Answer: B

6. Which built-in function creates a channel of integers?

A. channel(int)
B. new(chan int)
C. make(chan int)
D. chan.New(int)

Answer: C

7. What does a select statement do in Go?

A. Waits on several channel operations and runs the one that is ready
B. Chooses which goroutine the scheduler stops
C. Selects rows from a database table
D. Sorts the elements of a slice

Answer: A

8. Which command formats Go source code to the standard style?

A. go style
B. gopretty
C. gofmt
D. golint

Answer: C

9. A Go type satisfies an interface _____: it needs no implements declaration. [1 mark]

Answer: implicitly / automatically

10. In Go, the zero value of an int is _____ and the zero value of a string is the _____ string. [2 marks]

Answer: 0 / zero | empty

11. A Go module records its path and dependency versions in a file named _____. [1 mark]

Answer: go.mod

12. Explain the difference between an array and a slice in Go, including how each is passed to a function. [4 marks]

13. What is a data race? Describe how the Go race detector helps you find one. [4 marks]